

# HRISHABH MAHAJU

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## SUMMARY

Data Science M.S. graduate (UNH, May 2026) with 3+ years of production software engineering experience. Delivered measurable improvements in high-traffic systems across full-stack development, automation, cloud data pipelines, and AI agent development. Targeting Data Engineer, ML Engineer, or AI Developer roles.

## EDUCATION

**M.S. Data Science** - University of New Haven (Tagliatela College of Engineering), CT Aug 2024 – May 2026  
Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Distributed & Scalable Data Engineering (AWS), Power BI

**B.E. Computer Engineering** - Pokhara University (Cosmos College of Engineering), Nepal Sept 2016– July 2022  
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Web Technology, Artificial Intelligence, Distributed Systems

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, SQL

**ML / AI:** Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Keras, NLTK, OpenCV, Hugging Face, LangChain, RAG

**Web / Backend:** Spring Boot, Django, Flask, ReactJS, NextJS, Angular, NodeJS, REST APIs

**Cloud & Data:** AWS (S3, Kinesis, Lambda, Glue, Athena), Docker, Redis, MongoDB, MySQL, Firebase

**Tools:** Power BI, Git/GitHub, Selenium, JMeter, Postman, Burp Suite, JIRA, Scrum/Agile

## WORK EXPERIENCE

**Research Assistant** · Market Mind (AI-Driven Decision Support for Higher Ed Recruitment) · University of New Haven Aug 2025 – Present  
Technologies: Python, Scikit-Learn, Pandas, FastAPI, NextJS, TF-IDF/NLP | Advisors: Dr. Ardiana Sula, Dr. Muhammad Aminul Islam

- Co-developed an AI-driven MVP platform converting raw ad campaign data (Google, Meta, Bing, LinkedIn) and CRM records (9,000+ inquiries) into enrollment intelligence for university marketing teams.
- Built and validated 5 ML models (regression + classification) for ad performance prediction, content scoring, keyword ranking, and CRM funnel conversion (AUC up to 0.81).
- Led Phase 2 validation effort: reproduced Phase 1 benchmarks exactly, ran gap analysis on new campaign data, and authored data pipeline recommendations adopted for Phase 3 roadmap.

**AI Engineer Intern** · Webstars LLC / AI Trusted Advisors · Remote Dec 2025 – May 2026  
Technologies: Python, Voice AI, AI Agents, MCP, LangChain, ReactJS, NodeJS | Methodology: Agile

- Designing and developing custom Voice AI agents and MCP platform solutions to support client acquisition, internal automation, and partner integrations for a U.S.-focused AI advisory startup.
- Building backend infrastructure for agent/partner access portals, including usage tracking, billing, and platform reliability; responsible for weekly backups and outage response.
- Developing AI-powered cold outreach automation with A/B testing pipelines targeting 50,000–100,000+ emails/month, and building social media AI agents across LinkedIn, Reddit, Instagram, and TikTok.

**Software Engineer (SDET)** · YCO Pvt. Ltd. (Nepal Stock Exchange) · Kathmandu, Nepal Oct 2021 – Jun 2024  
Stack: Java, Spring Boot, Angular, Python, Selenium, JMeter, Redis, Docker, MSSQL, BIRT/Jasper | Methodology: Scrum

- Improved software performance by 50% through full automation testing (functional, regression, performance) across a microservices-based Fund Management and Trade Management System for Nepal's national stock exchange.
- Implemented High Availability (HA) multi-server architecture for nepalstock.com using Spring Boot, boosting system reliability and driving a 30% improvement in user engagement under peak-traffic conditions.
- Engineered Redis caching layer and optimized SQL stored procedures for high-throughput real-time data feeds, measurably reducing database load and API response times at national scale.
- Acted as Product Owner: owned QA backlog, prioritized features across 3 cross-functional teams, and conducted RCA investigations that cut recurring production incidents by ~40%.
- Containerized services with Docker; developed custom BIRT/Jasper reports replacing manual stakeholder reporting workflows; contributed to RBAC security refactoring for the exchange website.

## PROJECTS

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**Real-Time Traffic Analytics Pipeline** (*AWS Kinesis · S3 · Glue · Athena · Lambda*) — Cloud-native pipeline ingesting and analyzing simulated web traffic logs with end-to-end operational intelligence.

**DocuMate: AI PDF Assistant** (*Python · LangChain · FAISS · RAG · Streamlit*) — Conversational LLM assistant using RAG and vector-similarity retrieval to answer contextual questions from PDFs.

**Alzheimer's BRFSS Analysis** (*Python · Pandas · K-Means · Matplotlib*) — Trend analysis, confidence intervals, and clustering on CDC survey data to surface demographic health risk patterns.

**E-Health Care Chatbot** (*Python · Decision Tree · Flask*) — Symptom-based disease prediction chatbot with mental health consultation flow using a Decision Tree classifier.

## CERTIFICATIONS

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- MLOps Essentials: Model Deployment & Monitoring (LinkedIn Learning, Apr 2025) & MLOps Essentials: Model Development & Integration (LinkedIn Learning, Apr 2025)
- Apache Spark: Big Data Engineering (LinkedIn Learning, Apr 2025)
- Azure Spark Databricks (LinkedIn Learning, Apr 2025)
- Apache Kafka Essential Training (LinkedIn Learning, Mar 2025)
- SQL (Basic) – HackerRank Certified (Jan 2025)